

MACHINE LEARNING FUNDAMENTALS SERIES

Neural Networks Pt. 4: Full Backpropagation Details

Going Bonkers with the Chain Rule Across All 7 Network Parameters

Source: StatQuest with Josh Starmer — Backpropagation Details Pt. 2

1. Introduction and Core Idea

In previous parts, we explored Backpropagation by optimizing a single parameter (b_3) and then three output parameters (w_3, w_4, b_3). In this final section, we apply the **Chain Rule** and **Gradient Descent** to optimize **all 7 parameters simultaneously**:

$$\Theta = \{w_1, b_1, w_2, b_2, w_3, w_4, b_3\} \quad (1)$$

By calculating the partial derivative of the loss function (Sum of Squared Residuals, SSR) with respect to every weight and bias, Gradient Descent iteratively adjusts all 7 parameters so that the neural network's final output curve ("green squiggle") fits the training data accurately.

2. Neural Network Topology

Figure 1 illustrates the full 2-layer neural network architecture with 1 input x_i , 2 hidden Softplus nodes, and 1 output $\hat{y}_i$. All 7 trainable parameters Θ are highlighted in red.

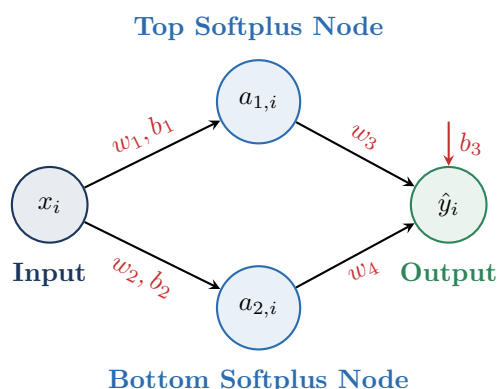


Figure 1: Full Neural Network Architecture highlighting all 7 parameters $\Theta = \{w_1, b_1, w_2, b_2, w_3, w_4, b_3\}$.

3. Full Chain Rule Derivations

The Loss function is the Sum of Squared Residuals (SSR) across n training samples:

$$\text{SSR} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

For the output layer parameters (w_3, w_4, b_3), the derivatives derived previously remain un-

changed:

$$\frac{\partial \text{SSR}}{\partial b_3} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \quad (3)$$

$$\frac{\partial \text{SSR}}{\partial w_3} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot a_{1,i} \quad (4)$$

$$\frac{\partial \text{SSR}}{\partial w_4} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot a_{2,i} \quad (5)$$

3.1. Chain Rule for Input Weight w_1 (4-Step Link)

To calculate the derivative of SSR with respect to the input weight w_1 of the top hidden node, we trace the path backward through 4 connected stages:

$$\text{SSR} \longrightarrow \hat{y}_i \longrightarrow a_{1,i} \longrightarrow x_{1,i} \longrightarrow w_1 \quad (6)$$

Applying the Chain Rule yields:

$$\frac{\partial \text{SSR}}{\partial w_1} = \sum_{i=1}^n \underbrace{\frac{\partial \text{SSR}}{\partial \hat{y}_i}}_{-2(y_i - \hat{y}_i)} \cdot \underbrace{\frac{\partial \hat{y}_i}{\partial a_{1,i}}}_{w_3} \cdot \underbrace{\frac{\partial a_{1,i}}{\partial x_{1,i}}}_{\frac{d}{dx} \text{Softplus}} \cdot \underbrace{\frac{\partial x_{1,i}}{\partial w_1}}_{x_i} \quad (7)$$

Evaluating each individual derivative step:

1. **Loss Derivative:** $\frac{\partial \text{SSR}}{\partial \hat{y}_i} = -2(y_i - \hat{y}_i)$.
2. **Output Path Derivative:** Since $\hat{y}_i = w_3 a_{1,i} + w_4 a_{2,i} + b_3$, we have $\frac{\partial \hat{y}_i}{\partial a_{1,i}} = w_3$.
3. **Activation Derivative:** $a_{1,i} = \ln(1 + e^{x_{1,i}})$. Taking the derivative with respect to $x_{1,i}$:

$$\frac{\partial a_{1,i}}{\partial x_{1,i}} = \frac{d}{dx} \ln(1 + e^{x_{1,i}}) = \frac{1}{1 + e^{x_{1,i}}} \cdot e^{x_{1,i}} = \frac{e^{x_{1,i}}}{1 + e^{x_{1,i}}} \quad (8)$$

4. **Linear Combination Derivative:** Since $x_{1,i} = w_1 x_i + b_1$, we have $\frac{\partial x_{1,i}}{\partial w_1} = x_i$.

Multiplying all 4 derivative terms together gives the Big Fancy Equation for w_1 :

$$\frac{\partial \text{SSR}}{\partial w_1} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot w_3 \cdot \left(\frac{e^{x_{1,i}}}{1 + e^{x_{1,i}}} \right) \cdot x_i \quad (9)$$

3.2. Chain Rule for Input Bias b_1

The path linking SSR to b_1 is identical to w_1 , except for the last term $\frac{\partial x_{1,i}}{\partial b_1} = 1$:

$$\frac{\partial \text{SSR}}{\partial b_1} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot w_3 \cdot \left(\frac{e^{x_{1,i}}}{1 + e^{x_{1,i}}} \right) \cdot 1 \quad (10)$$

3.3. Chain Rule for Bottom Node Parameters (w_2, b_2)

By symmetry, tracing the path through the bottom node ($a_{2,i}$) gives:

$$\frac{\partial \text{SSR}}{\partial w_2} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot w_4 \cdot \left(\frac{e^{x_{2,i}}}{1 + e^{x_{2,i}}} \right) \cdot x_i \quad (11)$$

$$\frac{\partial \text{SSR}}{\partial b_2} = -2 \sum_{i=1}^n (y_i - \hat{y}_i) \cdot w_4 \cdot \left(\frac{e^{x_{2,i}}}{1 + e^{x_{2,i}}} \right) \cdot 1 \quad (12)$$

4. Single-Iteration Gradient Descent Execution

Parameters are initialized prior to optimization:

- **Weights** (w_1, w_2, w_3, w_4) are initialized with random values drawn from a Standard Normal Distribution $\mathcal{N}(0, 1)$.
- **Biases** (b_1, b_2, b_3) are initialized to 0.0.

With a Learning Rate $\eta = 0.1$, we compute the gradients across all training samples in Iteration 1.

Table 1: Parameter Updates Across Iteration 1 of Simultaneous Backpropagation

Parameter	Initial Value $\theta^{(0)}$	Gradient $\frac{\partial SSR}{\partial \theta}$	Step Size ($\eta \cdot \text{Grad}$)	Updated Value $\theta^{(1)}$
w_1 (Top Weight)	-0.50	+0.76	+0.076	-0.576
b_1 (Top Bias)	0.00	+1.52	+0.152	-0.152
w_2 (Bottom Weight)	+0.80	-0.44	-0.044	+0.844
b_2 (Bottom Bias)	0.00	-0.88	-0.088	+0.088
w_3 (Top Output W)	+0.36	+0.22	+0.022	+0.338
w_4 (Bottom Output W)	+0.63	+0.96	+0.096	+0.534
b_3 (Output Bias)	0.00	+0.10	+0.010	-0.010

5. Convergence and Final Fit

After updating all 7 parameters simultaneously in Iteration 1, the network recalculates predictions $\hat{y}_i$ and enters Iteration 2.

Convergence to Optimal Parameters

As Gradient Descent repeats over **450 iterations**:

- The gradients $\frac{\partial SSR}{\partial \theta_j}$ for all 7 parameters asymptotically approach 0.
- The step sizes shrink towards zero, signaling convergence.
- The predicted output curve transforms into the final **”green squiggle”** that passes through all data points, achieving the global loss minimum.